Neighborhood operators

What Are Neighborhood Operators?

A neighborhood operator computes the output value of a pixel using the values of its neighboring pixels.

Unlike point operators (which use only one pixel), neighborhood operators consider local spatial context.

Mathematical Definition

For an image , a neighborhood operator produces:

Where:

- → neighborhood around pixel
- → operation (linear or non-linear)

Types of Neighborhoods

(a) 4-Neighborhood

(x,y−1)

(x−1,y) (x,y) (x+1,y)

(x,y+1)

Used for:

- Simple connectivity
- Low-complexity operations

(b) 8-Neighborhood

(x−1,y−1) (x,y−1) (x+1,y−1)

(x−1,y) (x,y) (x+1,y)

(x−1,y+1) (x,y+1) (x+1,y+1)

Used for:

- Smoothing
- Edge detection
- Texture analysis

1. Non-Linear Filtering

What Is Non-Linear Filtering?

A **non-linear filter** is a neighborhood operator where the output is **not a linear weighted sum** of pixel values.

Here, is a **non-linear function** such as:

- Median
- Minimum
- Maximum
- Rank selection

Why Non-Linear Filtering Is Used

Linear filters:

- Smooth noise
- But blur edges and fine details

Non-linear filters:

- Remove noise **without averaging across edges**
- Preserve important structures

Common Non-Linear Filters

1. Median Filter (Most Important)

Operation

- Replace the center pixel with the **median of neighborhood values**

Example (3×3 Neighborhood)

Example (3×3 Neighborhood)

Sorted values:

Median = 14

Advantages

Excellent for salt-and-pepper noise
Preserves edges
Simple to implement

Limitations

Not effective for Gaussian noise
Computationally expensive for large windows

Applications

- Medical image denoising
- Removal of impulse noise
- Preprocessing for segmentation

2. Bilateral Filtering

What Is Bilateral Filtering?

Bilateral filtering is a **non-linear, edge-preserving smoothing filter** that considers:

1. **Spatial closeness**
2. **Intensity similarity**

Unlike Gaussian filtering, it **does not blur edges**.

Mathematical Expression

Where:

- → spatial spread
- → intensity similarity
- → normalization factor

Why Bilateral Filtering Is Used

- Gaussian filter blurs edges
- Bilateral filter smooths **only similar pixels**
- Preserves object boundaries

Intuition

"Average only nearby pixels **that look similar**."

Advantages

✓ Preserves edges
✓ Reduces noise effectively
✓ Suitable for real-world images

Limitations

Computationally expensive
Parameter tuning is critical

Not suitable for real-time without optimization

Applications

- Image denoising
- HDR imaging
- Depth map smoothing
- Medical image enhancement
- Artistic image abstraction

3. Binary Image Processing

What Is Binary Image Processing?

Binary image processing deals with images having **only two pixel values**:

- 0 → background
- 1 → object

Binary images are usually obtained after **thresholding**.

Why Binary Image Processing Is Used

- Simplifies image analysis
- Focuses on **shape and structure**
- Reduces computational complexity

Basic Binary Neighborhood Operations

1. Logical Operations

- AND
- OR
- NOT
- XOR

2. Morphological Operations (Most Important)

These are **non-linear neighborhood operators** using a **structuring element**.

(a) Erosion

Removes boundary pixels

✓ Shrinks objects
✓ Removes noise

(b) Dilation

Adds pixels to boundaries

✓ Fills gaps
✓ Connects components

(c) Opening (Erosion → Dilation)

✓ Removes small objects
✓ Smooths contours

(d) Closing (Dilation → Erosion)

✓ Fills holes
✓ Bridges narrow gaps

Applications of Binary Image Processing

- Object detection
- OCR and document analysis
- Medical image segmentation
- Industrial inspection
- Fingerprint recognition